

Last name: _____
First name: _____
Matricola: _____
M.Sc. program: _____

Politecnico di Milano

Masters of Science in Computer Science and Engineering; Geoinformatics Engineering; Music and Acoustic Engineering; and Telecommunication Engineering

FOUNDATIONS OF OPERATIONS RESEARCH

Exam of February 11, 2022

Time available: 2 hours and 10 minutes

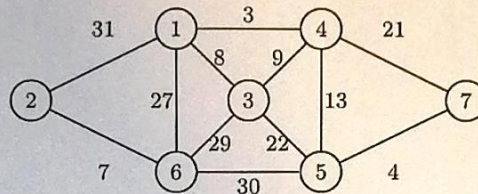
Prob. 1	Prob. 2	Prob. 3	Prob. 4	Prob. 5	Total
7.5	8.5	7.5	3.0	4.5	31

Remarks:

- Write the answers to the various questions in the specified locations using a blue or black pen.
- Report and explain all steps on the back of the previous page.
- The answers should be as clear and precise as possible.
- No course material or summary is allowed, except for Problem 5 which is open notes/book.

Problem 1

A company must interconnect seven new buildings via underground optical fiber cables in order to guarantee communications between every pair of buildings. Consider the following undirected graph whose nodes correspond to the buildings and whose edges correspond to the cables that can be installed between some pairs of buildings. Each edge is assigned a length expressed in multiples of 100 meters.



The goal is to determine how to interconnect all the buildings so as to minimize the total length of the cable required. For the sake of simplicity, we assume that the capacity of the cables to be installed is large enough to support any traffic between the seven buildings.

a) Provide a formal description of the graph optimization problem faced by the company, carefully specifying the inputs/outputs and precisely defining the feasible solutions.

b) Which algorithm can be used to solve the problem? Apply it to the above instance (reporting the intermediate steps on the back of the previous page) and indicate here the optimal solution found with the corresponding objective function value.

c) Consider the above optimal solution and determine for which (interval of) values of the length of the edge $\{5, 7\}$, currently equal to 400 meters, the optimal solution remains unchanged. Same question for the edge $\{3, 4\}$. In both cases, clearly explain why.

d) What is the computational complexity of the selected algorithm in terms of the number of nodes n and of edges m ? Motivate the answer.

Problem 2

Consider the linear programming problem:

$$\begin{array}{llllll} \min z = & 4x_1 & -x_2 & +x_3 & & \\ \text{s. t.} & x_1 & +2x_2 & -2x_3 & \leq & 2 \\ & 4x_1 & +3x_2 & +2x_3 & \geq & 6 \\ & x_1, & x_2, & x_3 & \geq & 0 \end{array}$$

a) Write the problem in the standard form.

b) State the Phase I auxiliary problem of the two-phase simplex method and mention the aim of this auxiliary problem.

c) Apply **one iteration** of the simplex method with Bland's rule starting from the initial basic feasible solution $\underline{x} = (3/2, 0, 0, 1/2, 0)^T$ associated with the following canonical form

$$\begin{array}{llllllll} \min z = & & 4x_2 & +x_3 & & -s_2 & +6 \\ \text{s. t.} & & 5/4x_2 & -5/2x_3 & +s_1 & +1/4s_2 & = & 1/2 \\ & x_1 & +3/4x_2 & +1/2x_3 & & -1/4s_2 & = & 3/2 \\ & x_1, & x_2, & x_3, & s_1, & s_2 & \geq & 0 \end{array}$$

Indicate here the resulting tableau and the corresponding basic feasible solution with its objective function value. Report on the back of the previous page the initial tableau and steps.

d) Write the dual of the original linear program.

e) What can you conclude about the optimal objective value of the dual problem from the answer in point c)? Clearly state the result that justifies your answer.

Problem 3

Consider the following Integer Linear Programming (ILP) problem:

$$\begin{array}{ll}\max & z = 2x_1 + 4x_2 \\ \text{s. t.} & 2x_1 + x_2 \leq 5 \\ & -4x_1 + 4x_2 \leq 5 \\ & x_1, x_2 \geq 0, \text{ integer}\end{array}$$

with the optimal tableau of the linear programming relaxation:

		x_1	x_2	x_3	x_4
	-25/2	0	0	- 2	-1/2
x_1	5/4	1	0	1/3	-1/12
x_2	5/2	0	1	1/3	1/6

a) Briefly describe the idea of the cutting plane method with Gomory cutting planes for ILP.

b) Draw the feasible region of the above ILP.

c) Determine the Gomory cutting plane associated with the basic variable with the smallest index.

d) Express the above Gomory cutting plane in terms of the decision variables x_1 and x_2 and draw it on the graphical representation of the feasible region of the ILP presented in point b).

e) Can we generate other Gomory cutting planes from the above optimal solution of the linear programming relaxation? Explain why and, if the answer is yes, exhibit another cut.

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Problem 4

Write the AMPL statements to model the following components:

- an integer parameter n ;
- a set $N = \{1, \dots, n\}$;
- a non negative real variable x_{ij} for all $i \in N$ and $j \in N$;
- a binary variable y_i for all $i \in N$;
- a parameter c_{ij} for all $i \in N$ and $j \in N$;
- a parameter a ;
- an integer parameter t .

Using the defined components, write the AMPL statements to model the objective function:

$$\min \sum_{i \in N} \sum_{j \in N} c_{ij} x_{ij}$$

and the following sets of constraints:

$$\sum_{j \in N: |c_{ij} - a| < \frac{a}{2}} x_{ij} + y_i \leq 1, \quad \forall i \in N$$

$$x_{ij} = 0, \quad \forall i \in N, \quad \forall j \in N : i \leq j + t.$$